🐄 Project Dora — Master Technical Blueprint

A Pixar-style interactive web animation: a virtual date proposal for Imane

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1. Folder Structure

```
dora-date/
├── index.html          ← The Stage
├── style.css           ← The Puppeteer
├── script.js           ← The Director
└── assets/
    ├── meadow.jpg      ← Your HD Swiss meadow background
    └── cow-dance.gif   ← Dancing cow GIF (see §9 for sourcing)
```

GitHub Pages deployment: Push the entire `dora-date/` folder as the repo root. Set Pages source to `/ (root)` → `index.html` will be the landing page. Share URL: `https://your-username.github.io/dora-date/`

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2. The State Machine (Single Source of Truth)

Dora lives in exactly ONE state at any time. `script.js` manages this by adding/removing a single class on `<div id="dora-wrapper">`.

```
INITIAL → state-walk → state-sit → state-idle → state-rizz
                                                    ↓
                                              [Sign appears]
                                            ↙            ↘
                                        YES              NO (×4 loop)
                                         ↓                ↓
                                  state-party         state-sad
```

State class	Trigger	Duration
`state-walk`	Auto at 1.5s (after curtain)	~3.5s
`state-sit`	Auto at 5.0s	Persistent
`state-idle`	Auto at 5.5s	~2.3s loop
`state-rizz`	Auto at 7.85s	Until choice
`state-sad`	On "No" click	Until YES
`state-party`	On "YES" click	Forever 🎉

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3. Act 1 — The Entrance (0s → 6s)

3a. Black Curtain

- Full-viewport `<div id="curtain">` with `background: #000`
- CSS transition: `opacity 1.5s ease-out` + `pointer-events: none` after fade
- Triggered: `setTimeout(() => curtain.style.opacity = 0, 300)`

3b. Background Stack (z-index layers)

```
z-index 0 : <img id="meadow-bg">  — object-fit: cover, full viewport
z-index 1 : <canvas id="dust-canvas">  — ambient golden particles, always running
z-index 2 : <div id="dora-wrapper">  — Dora's SVG
z-index 3 : <div id="sign-container">  — Wooden sign UI
z-index 4 : <canvas id="finale-canvas">  — Act 4 particle burst
z-index 5 : <div id="curtain">  — Black curtain (fades out)
```

3c. Dora's Walk (state-walk)

Dora starts at `left: -200px`, transitions to `left: 50%` + `transform: translateX(-50%)`.

Walk cycle (CSS keyframes on rigged SVG parts):

- `#dora-leg-fl`, `#dora-leg-bl` — front-left + back-left legs: `rotate(-20deg) → rotate(20deg)`, 0.45s, alternating phase
- `#dora-leg-fr`, `#dora-leg-br` — front-right + back-right legs: opposite phase

- `#dora-body` — subtle `translateY(0 → -4px → 0)` bob, 0.45s
- `#dora-tail` — lazy `rotate(-10deg → 15deg)` wag, 0.9s

3d. Dora Sits (state-sit)

- Walk cycle stops
- `#dora-body` rotates slightly forward (~8deg) and drops ~20px
- All four leg groups fold: legs animate to tucked position
- CSS `transition: all 0.6s cubic-bezier(0.34, 1.56, 0.64, 1)` (springy overshoot)

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4. Act 1 → Act 2 Bridge — The Pixar Beat (5.5s → 7.85s)

State: `state-idle` — Full Pixar Beat Sequence

Choreographed via staggered CSS animation-delay chains:

Time offset	Element	Animation
+0.0s	`#dora-body`	Subtle chest-rise breath (scale Y 1→1.02→1), 1.2s loop
+0.3s	`#dora-ear-left`	Single twitch: `rotate(-15deg)` snap then ease back
+0.7s	`#dora-ear-right`	Same twitch, slight delay
+1.0s	`#dora-head`	Slow `rotate(0 → 8deg)` head tilt right
+1.4s	`#dora-eye-left`, `#dora-eye-right`	Blink: scaleY(1 → 0.05 → 1), 0.18s
+1.8s	`#dora-head`	Hold tilt, slight curious bob

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5. Act 2 — Rizz & The Question (7.85s)

State: `state-rizz`

- `#dora-head` holds tilt from idle

- `#dora-blush-left`, `#dora-blush-right` — two SVG `<ellipse>` elements, opacity `0 → 0.85`, `transition: opacity 0.6s ease`
- Eyelash group `#dora-lashes` — `scaleY(1 → 1.4 → 1)` bat, twice

Wooden Sign Drop

- `<div id="sign">` is `position: absolute; top: -400px` initially
- A visual rope `<div id="rope">` is `position: absolute; top: 0; left: 50%`
- Drop animation: CSS `@keyframes signDrop` using `cubic-bezier(0.34, 1.2, 0.64, 1)` for spring feel
- On arrival: sign swings ±8deg with damped oscillation via chained keyframe steps
- Rope visual: thin `<div>` with `background: #5C3A1E`, `width: 4px`, `border-radius: 2px`

Sign Styling

```css
#sign {
  background: #6B3F1A;                          /* dark walnut */
  border: 6px solid #3D1F0A;
  border-radius: 4px;
  box-shadow: inset 0 0 20px rgba(0,0,0,0.4), 4px 4px 16px rgba(0,0,0,0.5);
  font-family: 'Amatic SC', cursive;            /* Google Fonts — hand-lettered feel *
  color: #F5E6C8;
  text-shadow: 1px 1px 2px rgba(0,0,0,0.6);
  padding: 24px 32px;
  max-width: 380px;
  text-align: center;
}
```

Buttons

- YES: `background: #2D7A3A`, large, `font-size: 1.3rem`, full warm green
- NO: `background: transparent`, `border: 2px solid #888`, small, `font-size: 0.9rem`

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6. Act 3 — The Rejection Loop

State: `state-sad` **(triggered on every "No" click)**

World effect: `#meadow-bg` gets CSS `filter: saturate(0.3) hue-rotate(200deg) brightness(0.8)` — cold blue desaturation. Transition: 0.8s ease.

Dora sad expressions:

- `#dora-eye-left`, `#dora-eye-right` — SVG paths morph to drooping arcs
- `#dora-ear-left`, `#dora-ear-right` — `rotate(20deg)` droop downward
- `#dora-head` — slumps forward `rotate(15deg)`
- `#dora-body` — `scaleY(0.92)` slight shrink
- Tear: `<circle id="dora-tear">` animates `cy` from eye downward, `opacity 1 → 0` at bottom

Button physics (JS-driven):

```js
noClickCount = 0; // tracked in script.js
// On each NO click:
noBtn.style.transform = `scale(${1 - noClickCount * 0.22})`;
yesBtn.style.transform = `scale(${1 + noClickCount * 0.18})`;
```

Text sequence:

- Click 1: "Wait... are you sure sure? 😦 "
- Click 2: "Think about the snacks though! 🍿 "
- Click 3: "Hchouma 3lik... you're breaking this cow's heart... 💔 "
- Click 4: NO button `display: none`. Text: "Okay, the NO button is tiny now anyway! 😂 "

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7. Act 4 — The Barnyard Finale

State: `state-party`

Triggered on YES click:

1. Sign container: `opacity → 0`, `translateY(-60px)`, then `display: none`
2. Cold filter removed: meadow returns to full color (0.5s transition)

3. `state-party` class applied to `#dora-wrapper`

Dora party animation (CSS keyframes):

- `#dora-body`: `translateY(0 → -30px → 0)` bounce, 0.5s, `animation-timing-function: cubic-bezier(0.34, 1.56, 0.64, 1)`, infinite
- `#dora-ear-left`, `#dora-ear-right`: frantic alternate `rotate(±25deg)`, 0.25s alternating
- `#dora-head`: side-to-side `rotate(-10deg → 10deg)`, 0.5s, infinite
- `#dora-body`: squash on landing (`scaleX(1.15) scaleY(0.85)`), stretch on air (`scaleX(0.9) scaleY(1.1)`) — classic squash-and-stretch

Dancing cow GIF:

```html
<img id="cow-gif" src="assets/cow-dance.gif"
     style="position:absolute; bottom: 80px; opacity:0;
            width: 180px; transition: opacity 0.5s;">
```

Fades in on YES. Positioned to one side of Dora so they "dance together."

Finale Canvas — Barnyard Physics: Spawns 60 particles on YES click. Each particle:

```js
{
  x, y,           // spawn at random x, top of screen
  vx, vy,         // random velocity
  emoji,          // random from ['🐄','🐮','⭐','✨','🎉']
  size,           // 24-48px
  rotation,
  rotationSpeed,
  bounce: 0.6     // coefficient of restitution
}
```

On floor contact: `vy *= -bounce`, apply squash (`scaleX: 1.3, scaleY: 0.7`) for 3 frames then restore — **squash-and-stretch**.

Text overlay:

```html
<div id="finale-text">Moo-yeah! Can't wait! ❤️</div>
```

Animates in with `scale(0.5 → 1.05 → 1)` + `opacity 0 → 1`, using `Amatic SC` at 3.5rem, warm golden color.

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8. Audio (Web Audio API — no external files needed)

All sounds are synthesized procedurally. No `.mp3` files required.

Sad Trombone (on "No" click)

```js
function playSadTrombone() {
  const ctx = new AudioContext();
  const osc = ctx.createOscillator();
  const gain = ctx.createGain();
  osc.connect(gain); gain.connect(ctx.destination);
  osc.frequency.setValueAtTime(400, ctx.currentTime);
  osc.frequency.exponentialRampToValueAtTime(180, ctx.currentTime + 0.8);
  gain.gain.setValueAtTime(0.3, ctx.currentTime);
  gain.gain.exponentialRampToValueAtTime(0.001, ctx.currentTime + 0.9);
  osc.start(); osc.stop(ctx.currentTime + 0.9);
}
```

Moo (on YES click)

```js
function playMoo() {
  const ctx = new AudioContext();
  const osc = ctx.createOscillator();
  const gain = ctx.createGain();
  osc.type = 'sawtooth';
  osc.connect(gain); gain.connect(ctx.destination);
  osc.frequency.setValueAtTime(150, ctx.currentTime);
  osc.frequency.setValueAtTime(180, ctx.currentTime + 0.15);
  osc.frequency.setValueAtTime(140, ctx.currentTime + 0.5);
  gain.gain.setValueAtTime(0.2, ctx.currentTime);
  gain.gain.exponentialRampToValueAtTime(0.001, ctx.currentTime + 0.9);
  osc.start(); osc.stop(ctx.currentTime + 0.9);
}
```

Note: Both fire only after a user gesture (click), which satisfies browser AudioContext autoplay policy.

9. The Dancing Cow GIF

Recommended search: Giphy → search `"cow dancing"` → filter by looping GIFs. Best candidates:

- **"Barnyard dance cow"** — matches the film aesthetic perfectly
- **"Happy cow bounce"** — simpler, still charming

Save as: `assets/cow-dance.gif`

If you can't find one you love, tell me and I'll use a CSS-animated SVG cow silhouette as a fallback — it'll look hand-crafted rather than borrowed.

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10. Dora's SVG Rig — Element IDs (The Full Anatomy)

Every animatable part has a predictable ID. `transform-origin` is set in CSS.

```
#dora-wrapper          ← outermost positioner (moves left→center in Act 1)
  #dora-shadow         ← soft ellipse on ground, scales with body
  #dora-body           ← torso (breathing, squash-stretch pivot)
    #dora-leg-bl       ← back-left leg  (transform-origin: top center)
    #dora-leg-br       ← back-right leg
    #dora-leg-fl       ← front-left leg
    #dora-leg-fr       ← front-right leg
    #dora-tail         ← tail (transform-origin: base of tail)
    #dora-head         ← head group (transform-origin: neck joint)
      #dora-ear-left   ← left ear  (transform-origin: base)
      #dora-ear-right  ← right ear
      #dora-eye-left   ← left eye (contains pupil + lid)
      #dora-eye-right  ← right eye
      #dora-lashes     ← lash group (scaleY for batting)
      #dora-nose       ← nose (static)
      #dora-mouth      ← mouth (path, morphs between states)
      #dora-blush-left ← left blush ellipse (opacity-animated)
      #dora-blush-right← right blush ellipse
      #dora-tear       ← animated tear circle (state-sad only)
    #dora-spots        ← decorative spots on body (static)
```

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11. Ambient Dust Canvas (Always Running)

`dust-canvas` runs a continuous RAF loop from page load.

Particle spec (50 particles):

```js
{
  x, y,           // random position
  vx: ±0.3,       // very slow drift
  vy: -0.2,       // gentle upward float
  size: 1.5–4,    // small
  opacity: 0–0.6,// semi-transparent
  color: '#FFD580' // warm gold
}
```

Particles that float off-screen top respawn at the bottom with new random x. Canvas uses `globalCompositeOperation = 'screen'` for a soft glow blend.

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12. Build Order (Session Plan)

We build in this exact order to always have a working file:

Session	Deliverable
1	`index.html` skeleton + `meadow.jpg` display + dust canvas running
2	Dora SVG rig drawn and placed on scene
3	Act 1: curtain lift + walk animation + sit
4	Act 1→2 bridge: idle Pixar beat
5	Act 2: rizz state + wooden sign drop
6	Act 3: rejection loop (sad state + button physics + text)
7	Act 4: party state + finale canvas + cow GIF + text overlay
8	Audio layer (trombone + moo)
9	Polish pass: timing tweaks, mobile viewport, GitHub deploy test

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13. Fonts

Load via Google Fonts in `<head>`:

```html
<link href="https://fonts.googleapis.com/css2?family=Amatic+SC:wght@400;700&display=
```

`Amatic SC` — hand-lettered, warm, quirky. Used on the sign text, buttons, and finale overlay. Perfect Pixar-storybook register.

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14. Critical Rules (Reminder)

1. **Never touch code outside the scope of the request.** Surgical edits only.
2. **No stiff fallbacks.** If an animation is complex, we keep it complex.
3. **Every edit specifies: file name + exact location.**
4. **The SVG rig is the single source of truth for Dora.** CSS animates it. JS never rewrites it.
5. **Canvas layers never overlap Dora's click targets.** `pointer-events: none` on both canvases.

Blueprint version 1.0 — ready to build.